

# Pavlo Melnyk

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## Education

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### PhD in Electrical Engineering with a specialisation in Computer Vision

Computer Vision and Learning Systems, Linköping University

Linköping, Sweden

Aug 2019 – Sept 2024

- Specialisation includes Machine Learning and Geometric Deep Learning
- Funded by [Wallenberg AI, Autonomous Systems and Software Program \(WASP\)](#) [↗](#)
- Under the supervision of Michael Felsberg at the [Computer Vision and Learning Systems](#) [↗](#) and WASP Graduate School
- 5-year degree with 30% of coursework (90 ECTS) and 20% of teaching and other academic duties
- Thesis “[Spherical Neuro\(n\)s for Geometric Deep Learning](#)” [↗](#)

### MEng in Computer Science and Technology

Hunan University

Changsha, China

Sept 2016 – June 2019

- Master’s thesis “Deep Learning for Offline Handwritten Chinese Character Recognition”

### BSc in Information Security Systems (Electrical Engineering)

Donetsk National Technical University

Pokrovsk, Ukraine

Sept 2012 – June 2016

## Experience

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### Postdoctoral Researcher

Computer Vision and Learning Systems, Linköping University

Linköping, Sweden

Feb 2025 -

- Research on geometry, including applications for materials science and human pose estimation

### Researcher in a Materials Science Project

Linköping University

Linköping, Sweden

Feb 2024 - Feb 2025

- WASP-WISE AI4Science collaborative project with Mårten Wadenbäck and Jonas Björk as PIs
- Developed an equivariant ML framework to be combined with DFT in a holistic approach enabling exploration of a broad range of materials and catalytic processes, resulting in the [CataLiUst](#) [↗](#) initiative

### Co-Founder, Co-Director

Demine Foundation

London, UK

Jan 2023 - Feb 2026

- Not-for-profit organisation developing ML-assisted tools for humanitarian demining
- ML development, data collection, and international collaborations

### Co-Founder, Board Member

Ukrainska Föreningen Östergötland (Regional Ukrainian Association) [ukrfo.se](#) [↗](#)

Linköping, Sweden

Mar 2022 -

- **Chairman** 2023 - 2025
- Association was awarded Vitsipps Prize 2023 for extraordinary civic efforts in Linköping Municipality

### Teaching Assistant, Supervisor of Master’s Theses

Computer Vision and Learning Systems, Linköping University

Linköping, Sweden

Aug 2019 -

- **1300+ hours of teaching** conducted in English and Swedish
- Laboratory exercises in Multidimensional Signal Analysis, Neural Networks and Deep Learning, and Computer Vision; lessons in Signal and Image Processing; projects in Computer Vision and Conceive-Design-Implement-Operate (CDIO)
- **23 Master’s theses** conducted at companies such as Maxar, Saab, Qualcomm, Ericsson, Bosch, Wikipedia, RISE, SICK, Combitech, etc.

### Doctoral Student

Computer Vision and Learning Systems, Linköping University

Linköping, Sweden

Aug 2019 - Sept 2024

### Student Representative

UNESCO Youth Forum

Changsha, China

May 2018

## Student Researcher

Key Laboratory of Embedded and Network Computing of Hunan Province

Changsha, China

Dec 2016 – June 2019

- Developed a SOTA [CNN-based method](#) [↗](#) for offline handwritten Chinese character recognition (3755 classes)

## Awards & Honors

- Challenge Winner, Swedish National Hackathon 2026
- WASP International Postdoctoral Grant (2 years), University of Pennsylvania (not taken up) 2025
- [Top Reviewer at NeurIPS 2023](#) [↗](#) (top 10%, 1,197 of 11,725 reviewers) 2023
- Honorable Mention in [ICML Topological Deep Learning Challenge](#) [↗](#) 2023
- WASP Doctoral Student Grant (5 years): Collaborative project "How to Inject Geometry into Deep Learning" 2019
- Award by Ministry of Science and Education of Ukraine (3 years): Recipient (1/50) of a Chinese Government Scholarship to pursue a Master's in China 2016
- Award by the Verkhovna Rada of Ukraine (1 year): Recipient of a two-term stipend as recognition of excellent achievements in studies 2014

## Reviewing Service

CVPR'22, 24, 26   NeurIPS'21, 23, 25   ICLR'24, 25   ICML'26   ICCV'25   ECCV'24   WACV'24   3DV'24

## Stack

Python · PyTorch · TensorFlow · NumPy/SciPy · scikit-learn · W&B · Git · Slurm

## Languages

**Ukrainian** — Native · **English** — Full professional proficiency · **Swedish** — C1 (advanced), certified (2021)  
**Chinese** — HSK5 (advanced), certified (2019) · **Spanish** — Basic

## Selected Publications

- [QuaMo: Quaternion Motions for Vision-based 3D Human Kinematics Capture](#) [↗](#) ICLR 2026  
Cuong Le, *Pavlo Melnyk*, Urs Waldmann, Mårten Wadenbäck, Bastian Wandt
- [Equivariant Modelling for Catalysis on 2D MXenes](#) [↗](#) EurIPS 2025  
Workshop on  
[SIMBIOCHEM](#) [↗](#)  
spotlight  
*Pavlo Melnyk*\*, Anmar Karmush\*, Ania Beatriz Rodríguez-Barrera, Mårten Wadenbäck,  
Michael Felsberg, Johanna Rosen, Jonas Björk\*
- [On Learning Deep  \$O\(n\)\$ -Equivariant Hyperspheres](#) [↗](#) ICML 2024  
*Pavlo Melnyk*, Michael Felsberg, Mårten Wadenbäck, Andreas Robinson, Cuong Le
- [TetraSphere: A Neural Descriptor for  \$O\(3\)\$ -Invariant Point Cloud Analysis](#) [↗](#) CVPR 2024  
*Pavlo Melnyk*, Andreas Robinson, Michael Felsberg, Mårten Wadenbäck
- [Steerable 3D Spherical Neurons](#) [↗](#) ICML 2022  
spotlight  
*Pavlo Melnyk*, Michael Felsberg, Mårten Wadenbäck
- [Embed Me If You Can: A Geometric Perceptron](#) [↗](#) ICCV 2021  
*Pavlo Melnyk*, Michael Felsberg, Mårten Wadenbäck
- [A High-Performance CNN Method for Offline Handwritten Chinese Character Recognition and Visualization](#) [↗](#) Soft Computing [↗](#)  
2020  
*Pavlo Melnyk*, Zhiqiang You, Keqin Li

See all publications on [Google Scholar](#) [↗](#)